P06

Laplace Domain Transfer Functions for Complex Linear Networks

Authors	H. W. Bode
Published	Network Analysis and Feedback Amplifier Design (1945)
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Domain	DOM_MATH

Abstract

By converting differential equations into rational functions H(s) = N(s)/D(s), systems are fully specified by their poles and zeros. Frequency response curves emerge directly on the imaginary s = j*omega axis.

Abstract quoted from H. W. Bode et al., Network Analysis and Feedback Amplifier Design (1945).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?